



Thor Fernholm

AI & ML Engineer

M.Sc in AI and Machine Learning

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Profile

Curious and driven AI and ML engineer with a foundation in full-stack software development. I have experience in building production-ready AI tools, including open-source RAG pipelines and tool-calling architectures for AI agents. While my technical interest spans multiple disciplines of AI and ML, I am especially excited about tackling the fundamental engineering challenges of building reliable AI agents.

Education

M.Sc in Computer Science and Engineering, Linköping University, Linköping

August 2019 - June 2023, August 2024 - June 2026

Five-year program combining a Bachelor's in Computer Science and Engineering with a Master's specialisation in AI and Machine Learning. Elective coursework includes deep learning, multi-agent systems, computer vision, and AI robotics. I gained deep foundational knowledge in computer systems and engineering principles, alongside the fundamental mathematics, statistics, and algorithms that form the backbone of AI.

Work Experience

Master's Thesis, Modular Finance, Stockholm

January 2026 - June 2026

For my master's thesis, I co-engineered a programmatic tool-calling (PTC) framework for a commercial AI agent. Collaborating as a two-person team, we were responsible for driving the entire project lifecycle, from engineering the system prototype to building custom evaluation pipelines. We used this setup to benchmark PTC against traditional in-context tool-calling, analyse performance trade-offs between task efficacy and operational efficiency, and correlate these metrics to human-perceived quality through user testing.

Software Engineer (AI), Modular Finance, Stockholm

June 2025 - August 2025

During this summer internship, I was responsible for engineering an end-to-end retrieval-augmented generation (RAG) pipeline tailored for processing financial data in production. I architected the solution as a reusable Go library to enable seamless integration into LLM-based agent workflows and other internal projects. My work resulted in an open-source release (github.com/modfin/ragnar), allowing instant RAG integration into any Go application.

Software Developer, Opter, Stockholm

September 2023 - August 2024

As a full-stack developer in the mobile app team, I was responsible for the continuous improvement and reliability of the Opter transport management system (TMS), a platform used by hundreds of logistics dispatchers across the Nordics. I collaborated within an agile team to engineer cross-platform mobile and desktop solutions with CI/CD practices using C# .NET and MAUI. Through this, I delivered continuous app upgrades and optimisations that significantly increased usability and user feedback.

Student Marketeer, Red Bull, Linköping

September 2022 - April 2023

I represented Red Bull as a brand ambassador, driving regional marketing initiatives. I was responsible for managing key reseller accounts, planning and executing promotional activities at large events, and engaging directly with consumers to strengthen brand presence and drive product awareness. In this role, I strengthened my communication skills and gained experience in stakeholder management.

Software Developer, Opter, Stockholm

Summer 2016, June 2017 - August 2019, Summer 2020

I was responsible for modernising legacy source code in the TMS by migrating and extending core functionality to a modern MVVM architecture. I also worked as a full-stack developer responsible for developing and maintaining the Windows Client in C# .NET and SQL. This included developing new features, bug resolution, and contributing to long-term system refinements. This experience taught me many core concepts of development, system architecture and operability, as well as how to collaborate efficiently in an agile environment.

Data Engineer Intern, Ericsson, Stockholm

June 2014 - July 2014

During my internship in the test department, I was responsible for processing and analysing large datasets from radio system tests to evaluate technical performance metrics and identify failure patterns to support engineering decisions. To optimise the workflow, I automated repetitive data manipulation tasks and generated data visualisations using Excel with custom scripts.

Voluntary Experience

Founder, Board Member, and Event Manager, Powerslide, Student Association, Linköping

April 2022 - June 2023

Board Member, Web, Photo, and Film, 720, Student Association, Linköping

April 2021 - June 2022

Public Relations and Event Manager, D-Group, Student Association, Linköping

March 2020 - June 2021

Skills

Programming Languages

Python, Golang, C#, C++, C, Java, R, GDScript, Matlab, Assembly

Frameworks

PyTorch, LangChain, TensorFlow, .NET, MAUI/Xamarin, ROS2

Databases and Search

SQL (PostgreSQL, MSSQL), Neo4j, pgvector, Meilisearch

Tools and Observability

Langfuse, OpenTelemetry, Git, IntelliJ Suite, Azure DevOps, Visual Studio, Godot

Languages

Swedish (Native), English (Fluent), Spanish (Beginner)